

Methods of Research Quiz (70 Points)

I. Multiple Choice (20 Points)

Choose the best answer.

1. Which characteristic of a good research means it starts and ends with a problem, leading to new problems?

A. Empirical

B. Logical

C. Cyclical

D. Analytical

2. Which purpose of research aims to find basic truths or principles, citing examples like Newton's Law or Freud's Theory?

A. Applied Research

B. Action Research

C. Descriptive Research

D. Pure Research

3. Which research method investigates a problem by studying the variables in retrospect, such as analyzing system failures?

A. Historical Research

B. Experimental Research

C. Descriptive Research

D. Ex Post Facto Research

4. In the context of the provided text, what is defined as the processed data that has meaning and use, such as a class average?

A. Raw Facts

B. Information

C. Records

D. Parameter

5. Which classification of data is gathered from published or second-hand sources like books, journals, and studies?

A. Primary Data

B. Secondary Data

C. Tertiary Data

D. Processed Data

6. The difference between the result from the sample and the true value of the population is termed as:

A. Parameter

B. Variable

C. Confidence Level

D. Sampling Error

7. Which type of non-probability sampling involves setting a fixed number of people from different groups, like 3 IT experts, 3 agency reps, and 4 users?

A. Accidental Sampling

B. Quota Sampling

C. Pure Random Sampling

D. Stratified Random Sampling

8. Which question type allows the respondent to choose more than one option?

A. Dichotomous

B. Multiple Choice

C. Recall Type

D. Multiple Response

9. Which element of a research problem refers to the place or locale where the study will be conducted?

A. Why

B. What

C. When

D. Where

10. Which characteristic requires that a Review of Related Literature (RRL) must be recent, usually covering the last 5-10 years?

- A. Objective and unbiased
- B. Reliable sources
- C. Relevant to the study

D. Recent

11. Which method of research includes the example "Survey on most-used apps"?

- A. Historical Research
- B. Experimental Research
- C. Ex Post Facto Research

D. Descriptive Research

12. A Capstone Project in IT focuses on which of the following areas?

- A. Giving a theory or partial solution
- B. Systematic investigation to solve a computing problem

C. Infrastructure, applications, or processes

- D. Discovering basic truths or principles

13. One of the reasons for needing sampling is that it:

- A. Consumes all sources of data

B. Helps save time and effort

- C. Makes research less accurate
- D. Avoids needing a parameter

14. Which sampling technique involves dividing the population into groups (strata) and then randomly picking from each group?

- A. Accidental Sampling
- B. Pure Random Sampling
- C. Quota Sampling

D. Stratified Random Sampling

15. What is the main problem cited with using Observation as a data gathering technique?

A. Some people are hard to reach.

B. It takes time and can be costly.

C. What you see might not always be true.

D. It is not good for those who cannot read.

16. Which type of interview is characterized by fixed questions with no changes allowed?

A. Non-standardized

B. Unstructured

C. Standardized/Structured

D. Guided-response

17. The section in Chapter 1 that discusses the Global and Local technology setting and the problems identified is the:

A. Objectives of the Study

B. Significance of the Study

C. Project Context

D. Scope and Limitation

18. Which source is explicitly listed as an INVALID SOURCE for a Review of Related Literature?

A. Research journals

B. Published theses

C. Legal documents

D. Wikipedia

19. Which citation style is primarily used in computer science, IT, and engineering?

A. APA

B. ACM

C. MLA

D. Chicago

20. In the ACM citation format, the reference list is ordered in what way?

A. Alphabetically by author

B. Alphabetically by title

C. In the order they appear in the text

D. By year of publication

II. Identification (15 Points)

Identify the term or concept being described.

21. A major advantage of mobile computing that allows users to work from anywhere. **Location Flexibility**
22. The characteristic of good research that requires precise judgment and confidence levels. **Critical**
23. The research method described as a systematic inquiry into an event to describe and explain a phenomenon (e.g., Company automation study). **Case Study Research**
24. The type of research that is decision-oriented and has immediate use, such as "Training students on online education." **Action Research**
25. The traditional work method, such as manual computation or preparing reports. **Manual Process**
26. The total group with common traits, like all students in a school. **Population**
27. The summary description of a variable in the whole population, like the average income of all families in a city. **Parameter**
28. The type of non-probability sampling where you ask whoever you meet by chance, with no system. **Accidental Sampling**
29. The data gathering technique that involves collecting data using questionnaires, which can be done personally, by phone, or online. **Survey**
30. The type of guided-response question where the respondent is asked to fill in details (e.g., Age, Sex). **Recall type**
31. The element of a research problem that defines the **subject or topic** to be studied. **What**
32. The specific objectives that must be **SMART** (Specific, Measurable, Attainable, Relevant, Time-bound). **Specific Objectives**
33. The meaning of a term based on its **dictionary meaning**. **Technical definition**
34. The purpose of this is to show where information came from, including the author, title, source, and date. **Citation**
35. The process of using someone else's idea but writing it in your own words instead of copying it directly. **Paraphrasing**

III. Modified True or False (30 Points)

Write **TRUE** if the statement is correct. If the statement is false, write **FALSE** and change the **underlined** word or phrase to make the statement true. (2 points each)

36. Research is analytical: it starts and ends with a problem, leading to new problems.

FALSE. Change analytical to Cyclical.

37. The example "Study of past ICT policies" is an example of Descriptive Research.

FALSE. Change Descriptive Research to Historical Research.

38. Information is raw facts, numbers, or records.

FALSE. Change Information to Data.

39. Primary Data is gathered from published or second-hand sources like books and journals.

FALSE. Change published or second-hand sources to original sources.

40. Sampling improves accuracy by focusing on a larger group.

FALSE. Change larger group to smaller group.

41. The research title must be written in inverted pyramid if more than one line.

TRUE.

42. Technical definition is the dictionary meaning of a term.

TRUE.

43. A good research problem should be within your skills and budget.

TRUE.

44. The problem with Interview as a data gathering technique is that some may not return forms or leave questions blank.

FALSE. Change Interview to Survey.

45. The What element of a research problem refers to the period or time data will be gathered.

FALSE. Change What to When.

46. The General Objective is one of the specific objectives that must be SMART.

FALSE. Change General Objective to Specific Objectives.

47. AI-generated references are considered valid sources for RRL.

FALSE. Change AI-generated references to Invalid sources (or a valid source like Research journals).

48. The APA format uses a numbered reference list, cited with brackets like [1].

FALSE. Change APA to ACM.

49. Paraphrasing helps you avoid plagiarism and keep your own voice.

TRUE.

50. You should use a Direct Quote when the author's exact words are not needed.

FALSE. Change Direct Quote to Paraphrase.

IV. Situational (Essay) (5 Points)

Scenario: Your research group decided to develop an "Automated Student Feedback System for Computer Science Department" and plans to submit the manuscript using the **ACM format**.

Based on the provided reviewer, answer the following:

1. Provide **three** criteria/guidelines for selecting a research problem/topic that justify the selection of this project. (3 points)
2. Explain why the choice of **ACM format** is appropriate for this specific research project. (2 points)

Answer:

1. Three Criteria/Guidelines Justifying the Topic Selection:

- **Related to your field:** Developing an "Automated Student Feedback System for Computer Science Department" aligns with the IT/Computer Science discipline.
- **Relevant and significant:** The study provides practical results by improving the feedback process for the school and students.
- **Researchable and manageable:** The scope of developing a "System" suggests it is a manageable project that can likely be finished on time.

2. Justification for using ACM format:

The choice of the ACM format is appropriate because the research is a computing-related study (an Automated System). The reviewer states that ACM is the citation and manuscript format specifically required and used in computer science, IT, and engineering disciplines.